databricks

# Spark at scale: Do's & Don'ts



Spark Do's & Don'ts workshop

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# Agenda

## Efficient large-scale computing on Databricks

- How Spark runs your code: driver, executors, lazy plans
- Where the time goes: shuffles, joins, data layout
- What serverless and Photon do for you, and what they can't
- Five ways to fight the engine, and what to do instead
- The workshop and the Genie Code challenge

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# Scale is a parallelism problem

## The same question, answered two ways

**Summing 100M order lines**

collect() to one Python process



≈ 13 min

aggregate in Spark, in parallel



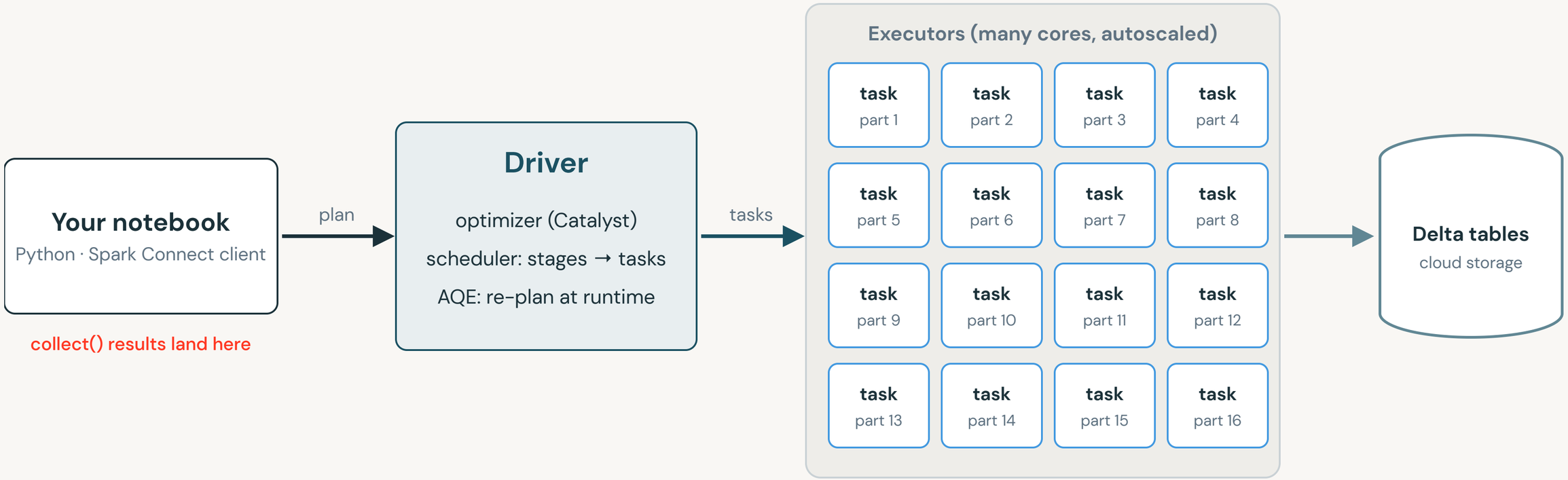

*Measured in the workshop on serverless (the collect() time is projected from a 4% slice).*

- Spark splits data into **partitions** and works on all of them at once, on many cores
- Efficient code keeps every core busy and **moves as little data as possible**

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# Anatomy of a Spark application

## On serverless, your notebook is a client and Databricks runs the engine




Only what you explicitly ask for comes back to the notebook. `collect()` and `toPandas()` ask for everything.

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# Lazy evaluation

## Transformations describe the work, actions run it




*transformations only build a plan: nothing has run yet*

**Transformations (lazy)**

`select · filter · withColumn · join · groupBy · orderBy · union`

**Actions (run the whole plan, every time)**

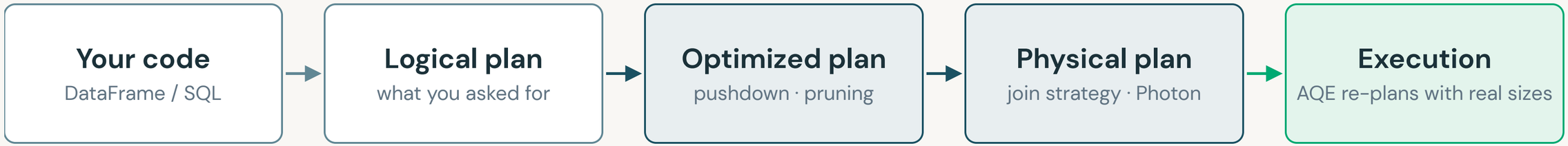
`count · collect · display · toPandas · write · isEmpty`

Five debug `count()` calls in a pipeline = the pipeline runs five times.

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# From DataFrame code to a physical plan

## The optimizer rewrites what you wrote, if it can see it




**The optimizer can rewrite**

– filters pushed down to the file scan

– only the columns you use are read

– small tables broadcast instead of shuffled

**...but it can't see inside**

– Python UDFs and pandas apply()

– lists you collect() and paste back in

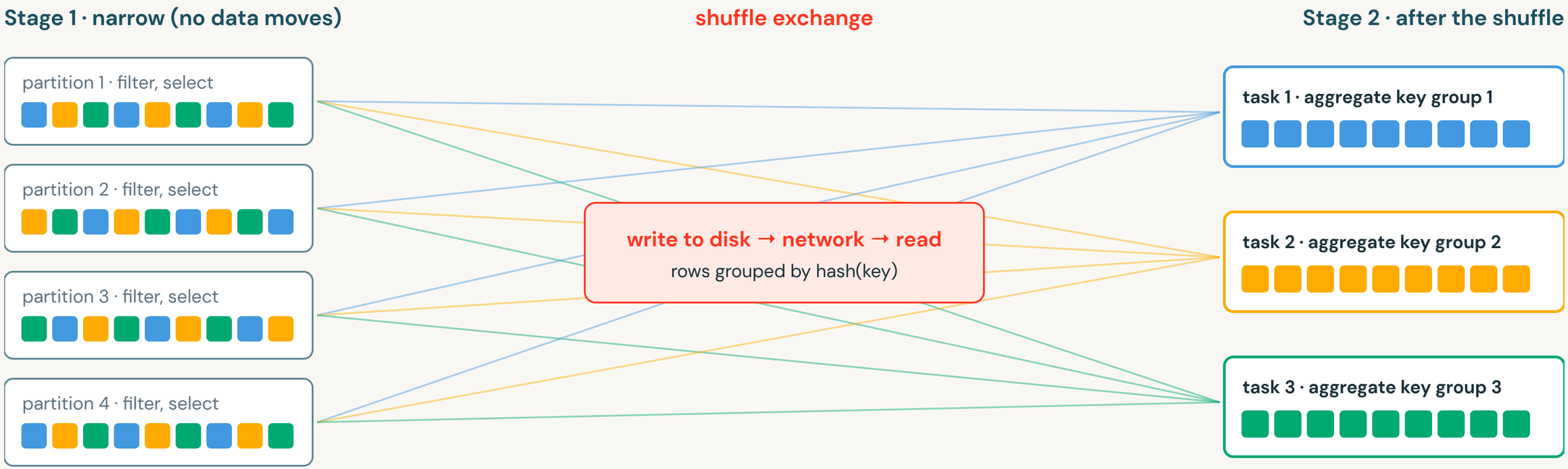
– loops that build 300-level-deep plans

Write code the optimizer can read: built-in functions, joins and filters, not Python black boxes.

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# Partitions, stages and the shuffle

## Narrow operations stream; wide operations move data



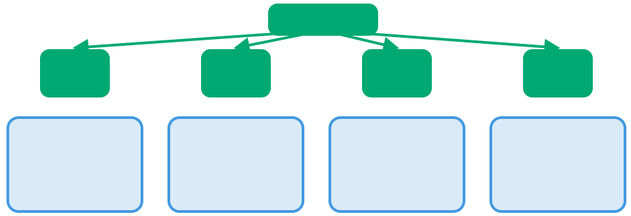

- **Narrow** (filter, select, withColumn): each partition on its own, no data moves
- **Wide** (groupBy, join, orderBy, window): rows with the same key must meet, so they cross the network
- One partition = one task = one core: a window with no `partitionBy` runs on a single core

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# How Spark joins

## The strategy, not the syntax, decides the cost

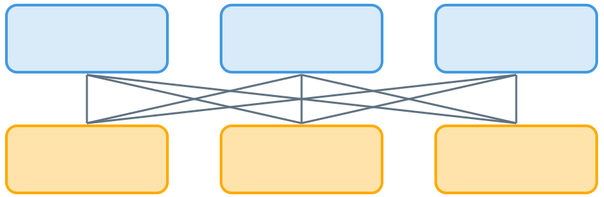
### Broadcast hash join



small side copied to every executor

**the big side never moves**

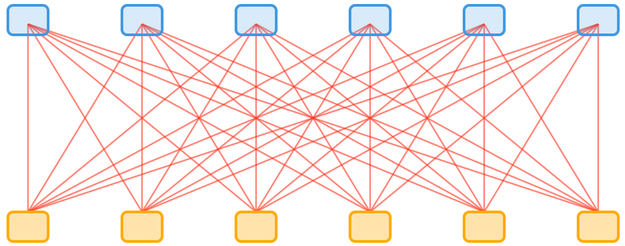
### Shuffle join



both sides re-partitioned by key

**fine for big ⋈ big, costs a shuffle**

### Nested loop / cartesian



no usable equality condition

**every row × every row**

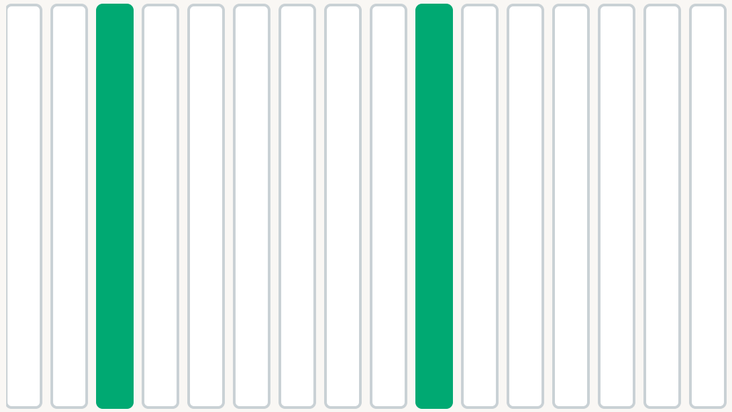
Joins on a non-unique key multiply rows. In the workshop, joining orders to an SCD2 history doubled revenue, with no error.

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# Data layout is performance

## The fastest data is the data you never read

### Columns: read only what you use



2 of 300 columns read from Parquet/Delta

toPandas() first = all 300 read and shipped

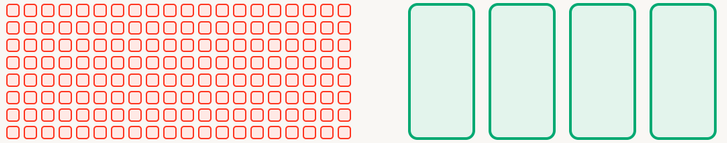
### Files: skip what can't match

- file 1 min/max: Jan–Feb
- file 2 min/max: Mar–Apr
- file 3 min/max: May–Jun
- file 4 min/max: Jul–Aug
- file 5 min/max: Sep–Oct
- **file 6 min/max: Nov–Dec**

`WHERE order_date >= '2025-11-01'`

liquid clustering keeps ranges tight

### Size: fewer, right-sized files




same rows: 6.7 s vs 1.1 s

OPTIMIZE / predictive optimization

aim for ~100 MB – 1 GB per file

Select the columns you need, filter on clustered columns, keep files right-sized: read less, finish sooner.

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# What serverless does for you

…and what is still up to your code

## The platform handles

- Adaptive query execution: broadcast decisions, partition sizes, skew
- Photon: vectorized, native execution of built-in operators
- Compute sizing and autoscaling
- Optimized writes, auto compaction, predictive optimization

## Your code still decides

- **Where** the work runs: executors or one driver process
- **Whether** the optimizer can see your intent
- **How often** the same work is repeated
- **How** data is laid out and keyed

```
spark.conf.set("spark.executor.memory", "16g")   # → [CONFIG_NOT_AVAILABLE] on serverless
```

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THE DO'S AND DON'TS

# Five ways to fight the engine

1. Keep the work on the executors
2. Let the optimizer see your intent
3. Don't do the work twice
4. Respect the data layout
5. Know your keys and your platform

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# 1 · Keep the work on the executors

The driver is one machine. Anything that funnels rows through it loses the parallelism you are paying for.

**DON'T**

- `collect()` / `toPandas()` on big DataFrames
- a Python loop running one query per key
- one `INSERT` per row
- `display()` left in production jobs

**DO**

- aggregate, filter and `limit` in Spark, then bring back the small result
- one `groupBy` or join over all keys
- one batch append, or `MERGE INTO`
- one action (the write); row counts from `DESCRIBE HISTORY`

collect() 100M rows (≈ 49 GB on a 15 GB driver)

**≈ 13 min → 1.6 s**

query per dealer ×500

**≈ 7 min → 0.8 s**

INSERT per row ×300

**≈ 6 min → 1.2 s**

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# 2 · Let the optimizer see your intent

Built-in functions and declarative joins can be optimized and run in Photon. Opaque Python and giant literal lists can't.

**DON'T**

- Python UDFs for arithmetic and if/else
- `withColumn` in a loop over hundreds of columns
- `isin(<100k collected IDs>)`
- `BETWEEN` joins with no usable equality

**DO**

- built-in functions (› pandas UDF › Python UDF)
- `withColumns({...})` or one `select`
- a left semi join
- the `range_join` hint, or a bucket key

Python UDF

**≈ 120 s → 1.4 s**

withColumn() loop

**≈ 83 s → 1.0 s**

range join

**33 s → 1.4 s**

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# 3 · Don't do the work twice

Every action re-runs the plan from the source. Every full reload re-reads all of history.

**DON'T**

- `count()` / `display()` after every step
- `if df.count() > 0`
- re-reading the whole landing zone each run

**DO**

- end the pipeline in one write; `cache()` only when you truly reuse
- `if not df.isEmpty()`
- Auto Loader or declarative pipelines (incremental)

debug actions per step

**31 s → 14 s**

count() > 0

**11.6 s → 0.7 s**

full reload vs Auto Loader

**14 s and growing → 8 s**

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# 4 · Respect the data layout

File count, file size and partitioning decide how much work every later read does.

**DON'T**

- thousands of tiny files
- `inferSchema` on CSV and JSON
- `partitionBy` on high-cardinality columns
- `coalesce(1)` before a big write

**DO**

- right-sized files; `OPTIMIZE` / predictive optimization
- an explicit schema, or Auto Loader schema hints
- liquid clustering (`CLUSTER BY`)
- parallel writes; aggregate first if you need one file

partitionBy(date, dealer)
**≈ 23 min → 13 s**

coalesce(1) write
**≈ 89 s → 12.5 s**

inferSchema
**17.6 s → 4.3 s**

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# 5 · Know your keys and your platform

Keys decide parallelism and correctness. Cluster-era tuning doesn't apply on serverless.

**DON'T**

- `Window.orderBy(...)` with no `partitionBy`
- joining on a key that isn't unique
- `spark.conf.set(...)` tuning, `sparkContext`, RDDs

**DO**

- `Window.partitionBy(key).orderBy(...)`
- one row per key: `is_current` or point-in-time
- DataFrame APIs, targeted hints; let AQE tune

window without partitionBy

**≈ 41 s → 7.4 s**

join on SCD2 history

**+100% revenue → exact**

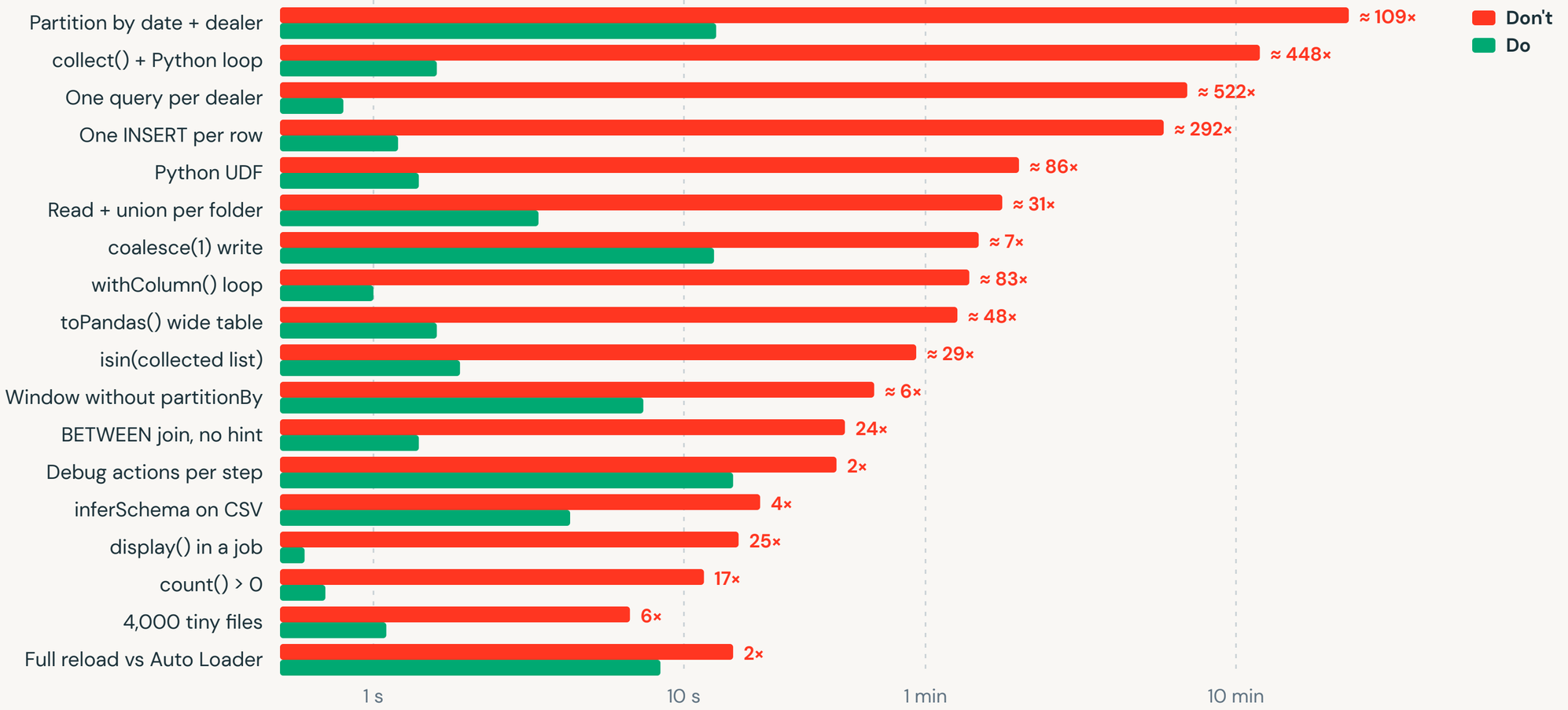
spark.executor.memory

**rejected → not needed**

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# Same answers, very different costs

## Every Don't/Do pair from the workshop, measured on serverless




*Don’t vs Do, log scale. ≈ marks a Don’t that ran on a slice of the workload and was projected to full size.*

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# The workshop

## Deploy it into your workspace as a Databricks Asset Bundle

```
databricks bundle deploy -t dev -p <profile> \
  --var="catalog=<catalog>" --var="schema=spark_dos_donts"

databricks bundle run -t dev -p <profile> \
  spark_dos_donts_setup   # ~5 min: 100M rows + raw files
```

- Open `01_spark_dos_and_donts` on serverless (env 5+)
- Each topic: **why**, then a timed **Don't** and **Do**
- Results land in a scoreboard table; every section can resume after a restart

### Genie Code challenge

A realistic dealer-scorecard job with **nine** anti-patterns from this deck. It gives the right answer, slowly.

- Ask Genie Code to find and fix them
- Paste the result; a check cell verifies the output
- Reference: ~2 min → **6 s**

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